

Name

[email] | [Number] | [LinkedIn](#) | [GitHub](#) | London/Reading, UK | Open to 2027 roles

EDUCATION

BSc Computer Science — University of Reading

Reading, UK

Expected Graduation: 2027

Predicted Classification: First Class

- Relevant Modules: Python Programming (80%), Data Structures & Algorithms (70%), Software Engineering (71%)

INDUSTRY EXPERIENCE

Software Developer Intern - Placement Year

Sep 2025 – Aug 2026

Elemental Concept 2016 Ltd, London

- Developed REST APIs and real-time WebSocket services in Python using FastAPI, implementing asynchronous request handling, structured validation, error handling, and multi-client broadcasting.
- Designed asynchronous data-access layers using SQLAlchemy 2.x and MySQL, applying Repository and Unit of Work patterns to improve maintainability, transaction control, and separation of concerns.
- Built containerised multi-service environments using Docker and Docker Compose, integrating Redis-backed sessions and caching alongside LocalStack-based DynamoDB and S3 workflows.
- Created automated unit and integration test suites using Pytest, validating API endpoints, persistence workflows, asynchronous services, and failure scenarios to reduce regression risk.
- Engineered and evaluated LLM backend services using FastAPI, Hugging Face Transformers, TruLens, and OpenAI LLM-as-a-Judge, reducing fallback usage from 30% to 17.5% and repair rates from 70% to 19% while maintaining 100% post-processed format validity.

Software Developer - Backend AI Intern

July 2026 – Sep 2026

FlyRank AI, Part-time, Remote

- Engineered an asynchronous AI job API using FastAPI, PostgreSQL, Redis, & Celery, implementing HTTP 202 polling, database-backed idempotency & atomic job claiming to prevent duplicate processing during task redelivery.
- Built a provider-neutral LLM integration with JSON Schema and Pydantic validation, bounded retries, timeouts, and exponential backoff, supported by 10 Pytest cases achieving 81% coverage across success and failure workflows.

TECHNICAL PROJECTS

Weather Intelligence Dashboard | *FastAPI, React, TypeScript, Redis, SQLite, SQLAlchemy, Pytest*

- Built a full-stack weather dashboard using FastAPI, React & TypeScript, enabling users to retrieve live conditions & 5-day forecasts, use browser geolocation, and manage persisted weather records through a responsive interface.
- Designed a layered REST API using FastAPI, Pydantic, SQLAlchemy & SQLite, implementing validated CRUD operations across route, service, repository & database layers to improve maintainability & separation of concerns.
- Integrated WeatherAPI with Redis-based caching, configurable TTLs, normalised cache keys, and fail-open fallback behaviour, reducing average repeated-request latency from 219 ms to 23 ms in a 20-request local benchmark, an 89.5% reduction and 9.5× speed-up.

JavaFX Matrix Solver | *Java, JavaFX, Maven, JUnit*

- Developed a JavaFX matrix-solving application using the MVC design pattern, implementing inversion, determinants, row reduction, systems-of-equations solving, and a custom mathematical expression parser.
- Engineered input validation, exception handling, and 50+ JUnit tests covering core matrix operations and edge cases, improving application robustness and reducing regression risk.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C/C++

Backend & APIs: FastAPI, Flask, SQLAlchemy 2.x, Celery, REST APIs, WebSockets, Pydantic

Databases & Caching: PostgreSQL, MySQL, SQLite, DynamoDB, Redis

Cloud & DevOps: AWS (S3, DynamoDB), Docker, Docker Compose, LocalStack, Linux

Testing & Developer Tools: Pytest, JUnit, Git, Maven

CERTIFICATIONS & AWARDS

RED Outstanding Achievement - Professional Development, Global Engagement and Sustainable Action

MITx Courses - Machine Learning with Python: from Linear Models to Deep Learning

Anthropic - Claude with Amazon Bedrock